Instructor answer key and scoring

Scoring per team: +10 per hazard found (right grid cell or right spot), +5 if the type is also right, -5 per false flag (flagging a correct practice or empty ground). Max 16 x 15 = 240. Optional: +20 to the first team to finish, +10 to the second. Tie-break: fewer false flags.

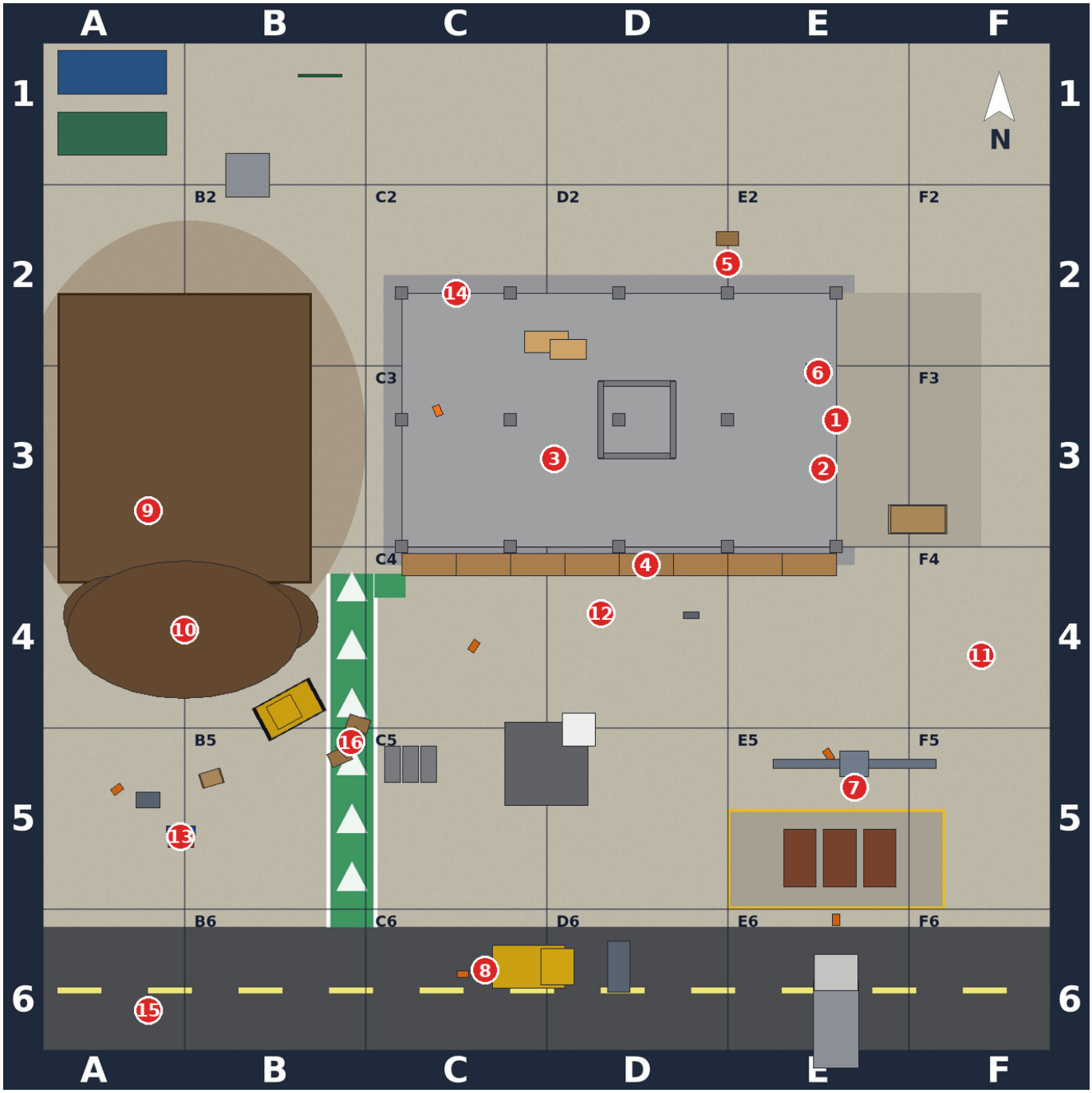
#	Cell	Type	Hazard and what to look for	Correct control
1	E3	Fall	Open slab edge (no guardrail). Top slab: the east edge has no guardrail or toe-board, while the other three edges are protected.	Install top rail, mid rail and toe-board before anyone works near the edge.
2	E3	Fall	Worker at the edge, not clipped on. Worker stands right at the unprotected east edge with no harness or lanyard.	Fit edge protection first; if work at the edge is unavoidable, harness tied to an anchor point.
3	D3	Fall	Uncovered floor opening. Hole in the level-2 slab with no cover, rail or warning. (The orange panel on the same level is a properly covered opening.)	Cover with a secured, marked cover (or fit guardrail around it).
4	D4	Fall	Scaffold platform gap, no edge protection. Top scaffold lift, bay 5: half the boards are missing, a board hangs loose, and guardrails and toe-board are missing.	Complete boards, guardrails and toe-boards; tag the scaffold "do not use" until fixed.
5	E2	Fall	Unsecured, badly placed ladder. Ladder on the north face is too flat, stands on a pallet, is not tied, and does not extend above the landing.	Ladder at 4:1, on firm level ground, tied off, extending 1 m above the landing.
6	E3	Struck-by	Material stacked on the slab edge. Pallet of bricks on the open slab edge: it can fall onto people below.	Keep stacks away from edges; use toe-boards; keep the area below barricaded.
7	E5	Struck-by	Crane load over workers. Suspended steel load is slewed directly over two workers; there is no exclusion zone, tag line or signaller.	Never work under a load; barricade the lift zone and use a signaller.
8	C6	Struck-by	Reversing loader, person in blind spot. Wheel loader is reversing with a worker directly behind it and no banksman. (The dump truck at E6 has a banksman and barriers.)	Separate people and plant; use a banksman; never stand in a machine blind spot.
9	A3	Caught-in/between	Unshored excavation with a worker inside. South half of the pit has vertical walls with no shoring, no barrier and no ladder, and a worker is inside. (The north half is shored, barricaded and has a ladder.)	Shore or batter the sides, provide a ladder, barricade the edge.
10	B4	Caught-in/between	Spoil heap at the excavation edge. Excavated soil is piled on the edge of the pit, adding load and risking a collapse onto the worker.	Keep spoil at least the depth of the excavation (or more) away from the edge.
11	F4	Electrical	Live overhead line, no warning or guard. Overhead power line runs through the crane and plant working area with no marker, goal-posts or isolation.	Ask the utility to isolate or divert; set exclusion distance, goal-posts and warning signs.
12	D4	Electrical	Damaged cable in a puddle, open board. Cable lies across the walkway in a puddle, one end is stripped, and the distribution board door is open.	Replace the cable, lift it off the ground, close/lock the board, use RCD protection.
13	A5	Fire/Explosion	Loose gas cylinders near welding. Oxygen and acetylene cylinders lie on the ground, not secured, next to welding sparks and a wooden pallet stack, with no extinguisher. (The caged, chained cylinders near the cabin are correct.)	Store upright, chained, apart; keep combustibles away from hot work; extinguisher at hand.
14	C2	Housekeeping/Other	Uncapped rebar ends. Bare rebar sticking up from the column stubs at the north-west corner of the top slab. (The yellow-capped bars at the next columns are protected.)	Fit rebar caps (or bend over / cover) so a fall or trip cannot impale a worker.
15	A6	PPE	No helmet or hi-vis on a live road. Worker walks along the site road without a helmet or high-visibility clothing.	Mandatory PPE for everyone on site; enforce at the gate.
16	B5	Housekeeping/Other	Debris blocking the walkway. Pallets, pipes and offcuts block the marked pedestrian route and the access to the building.	Keep walkways clear; designated storage; regular housekeeping checks.

Decoys: things done correctly (flagging these is a false flag)

- Top-slab edges with guardrail: West, south and north edges of the top level are fully protected.
- Worker clipped on behind a railed edge: Harness and lanyard tied to an anchor post: correct practice.
- Shored, barricaded excavation (north half): Sheeting, struts, ladder and barrier: correct practice.
- Caged, chained gas cylinders at the cabin: Upright, secured, extinguisher beside them.
- Dump truck with banksman and barriers: Signaller guides the reversing truck; area is barricaded.

- Covered, marked floor opening: Orange cover with fixings on the level-2 slab.

Answer map


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C6
D6
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